

NAME

ioperm – set port input/output permissions

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <sys/io.h>
```

```
int ioperm(unsigned long from, unsigned long num, int turn_on);
```

DESCRIPTION

ioperm() sets the port access permission bits for the calling thread for *num* bits starting from port address *from*. If *turn_on* is nonzero, then permission for the specified bits is enabled; otherwise it is disabled. If *turn_on* is nonzero, the calling thread must be privileged (**CAP_SYS_RAWIO**).

Before Linux 2.6.8, only the first 0x3ff I/O ports could be specified in this manner. For more ports, the **iopl(2)** system call had to be used (with a *level* argument of 3). Since Linux 2.6.8, 65,536 I/O ports can be specified.

Permissions are inherited by the child created by **fork(2)** (but see NOTES). Permissions are preserved across **execve(2)**; this is useful for giving port access permissions to unprivileged programs.

This call is mostly for the i386 architecture. On many other architectures it does not exist or will always return an error.

RETURN VALUE

On success, zero is returned. On error, -1 is returned, and *errno* is set to indicate the error.

ERRORS**EINVAL**

Invalid values for *from* or *num*.

EIO (on PowerPC) This call is not supported.

ENOMEM

Out of memory.

EPERM

The calling thread has insufficient privilege.

VERSIONS

glibc has an **ioperm()** prototype both in *<sys/io.h>* and in *<sys/perm.h>*. A void the latter, it is available on i386 only.

STANDARDS

Linux.

HISTORY

Before Linux 2.4, permissions were not inherited by a child created by **fork(2)**.

NOTES

The */proc/ioproports* file shows the I/O ports that are currently allocated on the system.

SEE ALSO

iopl(2), **outb(2)**, **capabilities(7)**